LIDA SHAHMIRI

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Professional Summary

PhD in Computer Science specialising in computer vision and machine learning, with expertise in deep learning, explainable AI, and dataset optimisation for robust image classification systems. Experienced in developing and evaluating deep learning models for classification and segmentation, with particular emphasis on dataset selection, model generalisation, and validation. Proficient in Python and modern ML frameworks, with practical experience building experimental ML pipelines and prototype applications. Comfortable working in interdisciplinary environments and communicating technical ideas clearly.

Skills

- **Programming:** Python (OpenCV, PyTorch, Scikit-Learn, TensorFlow, Keras), C#, SQL
- **Statistics & ML:** Statistical modelling, hypothesis testing, regression, classification, clustering, model evaluation, explainable AI (LIME/SHAP/Grad-CAM), LLMs, RAG.
- **Visualisation & Tools:** Tableau, Power BI, Matplotlib, Seaborn, Streamlit, GitHub, GitHub Actions, Docker, LaTeX, APIs
- **Cloud Platforms:** AWS (S3, EC2, RDS, SageMaker, Lambda), Azure
- **Soft Skills**: Analytical Thinking, Problem-Solving, Team Collaboration, Adaptability & Quick Learning

Education

PhD in Computer Science| Open University | UK **Feb 2021- Sep 2025**

- Design a robust framework for the classification of medicinal plants in the natural environment
- Proposed two novel dataset partitioning methods (MIGT and GMM-MIGT) for CNN training, reducing misclassification rates by up to 70% (🔗 migt).
- Introduced a quantitative analysis of the LIME technique for model validation

MSc in Computer Science | Islamic Azad University | Iran **Feb 2013- Sep 2015**

BSc in Computer Science |Payam Noor | Iran **Feb 2007- Jan 2011**

Experience

Research Assistant | The Open University| UK **Mar 2026- Present**

- Analyse and integrate complex longitudinal datasets, including data cleaning, preprocessing, feature engineering, and preparation of analysis-ready datasets from multiple sources.
- Develop and evaluate machine learning and deep learning models, to identify patterns and trends in data.
- Apply explainable AI and data visualisation techniques and document methodologies, experiments, and results in technical reports.

Research Assistant | The Open University| UK **Jan 2026- Feb 2026**

- Contributed to the UK Deepfake Detection Challenge, with involvement beginning prior to formal contract
- Played a key role in explainable AI for deepfake image detection, applying advanced image analysis techniques to improve model interpretability and localisation.

Achievements:

- 1st place – Explainability and Localisation category
- 1st place – Academic category
- 7th place – Overall
- The proposed model was commended for its strong performance in detecting deepfakes generated by the latest AI techniques, as well as for the clarity and usefulness of its explanations.

Research Assistant | The Open University| UK **Apr 2025- Jul 2025**

- Developed Zooniverse butterfly wing pattern recognition for citizen science.
- Applied image processing techniques like U-Net segmentation to improve landmark visibility and reduce background noise.
- Conducted research on butterfly wing pattern recognition
- Supported the successful acquisition of a research grant at The Open University

Researcher & Developer |Pooya Samaneh Diva |Iran **Feb 2018- Jan 2020**

- Supervised the development of novel software products for the Hospital Information System (HIS), in accordance with health specifications and security of information.
- Reviewed experimental results and user feedback to establish system functionalities and streamline patient management workflows.

Personal Projects

Creating An Image Search Engine Using DL - 🔗 View blog post | View code on GitHub | App
Python | Flask | TensorFlow | Keras | VGG16 | Cosine Similarity | KNN | Docker | Railway

- Developed a deep learning–based image search engine that retrieves visually similar products by comparing image embeddings.
- Extracted deep visual features using a pre-trained VGG16 model with Global Average Pooling for compact and efficient representation.
- Implemented Cosine Similarity and K-Nearest Neighbours (KNN) to identify and rank the most visually similar items.
- Built a Flask web interface enabling users to browse product images and instantly view visually similar matches.
- Containerised and deployed the application using Docker on Railway for scalable, production-ready cloud performance.

AI-Powered SQL Agent 🔗 View blog post | View code on GitHub | App
Python | Flask | LangChain | OpenAI GPT-4.1 | PostgreSQL | SQLAlchemy | Render Cloud

- Built an AI-driven SQL Agent that converts natural-language questions into optimised and safe PostgreSQL SELECT queries.
- Connected the agent to a real PostgreSQL database and automated schema discovery, table understanding, and query validation.
- Developed a Flask web interface allowing users to submit questions and receive both the SQL query and a human-readable explanation.
- Deployed the full application to Render Cloud, enabling real-time interaction with the live database.

Professional Highlights

- Won, UK Deepfake Detection Challenge. 1st place in Explainability & Localisation, 1st in Academic category, and 7th overall **2026**
- Won grant funding in a collaborative research project: Butterfly Wing Diversification & Evolution with Citizen Science, The Open University, Research assistant / Co-Applicant **2025**
- Published three first author and one co-author peer-reviewed articles **2019- 2025**
- Summer digital workshop AI Lead, Fresh Youth, Milton Keynes, UK **2025**
- Presented talks at conferences: C&C, poster competition (UK), IAU (Iran), IPAS (Italy) **2021- 2025**
- PhD scholarship **2007, 2021**
- Organising workshops and seminars and giving lectures at two universities in Iran **2015- 2018**
- Master's degree, Graduated with Distinction **2015**